

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, SQL

Deep Learning & ML: PyTorch, TensorFlow, Deep Neural Networks, CNN, RNN, Transformers, Perception, Segmentation, Prediction, Planning, Time Series Prediction, Active Learning, Reinforcement Learning, Data Science Methods

MLOps & Deployment: Docker, Kubernetes, AWS SageMaker, FastAPI, Git, CI/CD, Agile Development, Real-time Systems, Embedded AI, TensorRT

Data Engineering: PySpark, Airflow, ETL Pipelines, Data Handling

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated **deep learning models** for metrological applications, enhancing **model performance** and training efficiency.
- Explored **novel architectures** and physics-informed hybrid loss functions to improve model accuracy and stability, applying **state-of-the-art ML techniques**.
- Collaborated with PhD researchers on a publication for FLOMEKO conference, demonstrating a **scientific background** in relevant topics.
- Built data dashboards for model performance monitoring, emphasizing **data science methods** and a **systematic working style**.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Enhanced **vitiligo segmentation** accuracy by 94% and inference speed by 5x using **deep neural networks** and **AI algorithms**, deployed with **Docker** and **FastAPI**.
- Engineered a **real-time perception** system for tennis foul detection using **TensorRT**, reducing compute cost and processing time by 6x, showcasing **embedded AI** capabilities.
- Managed **MLOps pipelines** on **AWS SageMaker**, addressing **YOLO model** data drift, and maintained **Kubernetes infrastructure**, adhering to **agile development** processes.
- Developed **Proof of Concepts (POCs)** for **perception** tasks using **YOLO**, **few-shot learning**, and **CNN**, integrating **state-of-the-art ML techniques** for client solutions.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed **AI-based services**, including a chatbot application, leveraging **data-driven approaches** and modern **ML frameworks** (OpenAI API).
- Applied **Python** for full-stack development, delivering robust and scalable **AI** solutions.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Engineered **ETL pipelines** for robust **data handling** and automated data ingestion/cleaning workflows, supporting **data science methods**.
- Utilized **Python** to develop scalable web scrapers and data processing solutions for diverse client analytics needs.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, using AlphaFold for feature extraction, demonstrating advanced **ML techniques** and **analytical skills**.
- Increased **model accuracy** by 4% through custom **feature engineering**, applying a **systematic** approach to complex biological **prediction** tasks.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a thesis project focused on **real-time Bitcoin price forecasting**, implementing **time series prediction with modern ML techniques**.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Designed a **real-time signal processing** system for music generation, integrating temporal dependencies and **transformer-based models** for **sequence prediction**.
- Collaborated on the world's first AI-driven self-playing Chitralli Sitar, showcasing application of **modern ML techniques** in complex, interactive systems.

Hugging Face

Open-Source

Contributor

- Co-authored technical documentation and implementation for **Zero-shot Image Classification** on the Hugging Face Hub, demonstrating expertise in **deep neural networks** and **perception**.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in **deep learning** for **computer vision** and spatio-temporal modeling, relevant to **perception** and **prediction** in automotive applications.
- Relevant coursework: Advanced Neural Networks, Computer Vision, and **Autonomous Systems**.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python